# Random-Subspace Sweep Tests: Multiplicity-Sensitive Decay, Physical Error Budgets, and Exception-Count Compression

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**Abstract**

The assigned coordinate-sweep manuscript obtains a full-permutation mixing assertion from a uniform regular Schatten moment, using random-subspace retention, orthogonal cell tags, and an induction over balanced coordinate splits. Its nonnormal power conversion is already correct: it uses Schatten Hölder, not an equality between powers of a Gram matrix and Gram matrices of powers. We preserve that distinction and the unnormalized regular trace. We derive an explicit carrier-multiplicity gain for longer and heterogeneous sweeps, an original-law perturbation budget for biased matching layers after a genuinely smoothed prefix, and a smaller finite cell-tag bound obtained by retaining orthogonality between different exception counts. Physical Gram transport and representation-tail obligations are stated explicitly. The refinements are elementary, conditional improvements of finite estimates, not a new optimal-order shuffle theorem, an explicit universal sweep count, or an independent verification of the imported routing theorem. Contribution assessment: C2/5, modest source-specific multiplicity, biased-layer, and exception-count refinements using classical norm tools, without new global mixing or complexity claims.

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# 1 Model, status, and proof boundaries

Put $n = 2^d$, $d \geq 1$, and label positions by $\mathbb{F}_2^d$. A physical Thorp step rotates the coordinates and adds an independent fair bit to the departing coordinate, separately for each of the $n/2$ pairs. In the rotating frame this is one coordinate matching. After $d$ physical steps the rotation returns to the identity and one complete coordinate sweep has occurred. The full state space is $G = S_n$, not the $n$-point one-card space.

Let $H_i \cong (\mathbb{Z}/2\mathbb{Z})^{n/2}$ be the subgroup generated by the disjoint transpositions in coordinate $i$. In a unitary representation $\rho$,

$$P_i = \frac{1}{|H_i|} \sum_{h \in H_i} \rho(h), \qquad P_i = P_i^* = P_i^2, \qquad Q = P_d \cdots P_1. \tag{1}$$

The rightmost factor acts first. All coins in separate layers and sweeps are independent. A prescribed permutation of the coordinate order is allowed below, but deck-dependent scheduling is not silently identified with an independent convolution law.

We use counting-measure $\ell^2(G)$; a uniform rescaling of its inner product does not change operator singular values. Write $N = |G| = n!$, $\Pi = N^{-1}\mathbf{1}\mathbf{1}^*$, and $A = Q - \Pi$. All convolution operators commute with $\Pi$. For $p > 0$, $\|A\|_{S_p}^p = \operatorname{Tr}|A|^p$, with an *unnormalized* trace. The regular decomposition is

$$\ell^2(G) \simeq \bigoplus_{\lambda \vdash n} V_\lambda \otimes \mathbb{C}^{D_\lambda}, \qquad Q \simeq \bigoplus_{\lambda \vdash n} Q_\lambda \otimes I_{D_\lambda}. \tag{2}$$

Thus $D_\lambda$, rather than one, is the multiplicity of each carrier singular value. The trivial carrier contributes exactly one.

The assigned source [1] asserts that one absolute $P \geq 2$ works for all $d \geq 1$:

$$\operatorname{Tr}_{\mathrm{reg}} |Q|^P \leq 1 + n^{-10}. \tag{3}$$

This report treats (3) as an imported theorem when deducing mixing consequences. Reading its proof is not a substitute for verifying all companion routing and representation-theoretic inputs. In particular, we do not compute its unspecified universal constant $P$.

## 1.1 Current public context

Morris's published 2013 result gives $O(d^3)$ physical shuffles on $2^d$ cards; the primary arXiv version is dated December 14, 2009 [3]. We refreshed the primary abstract and publisher metadata, not that complete proof. The publisher's current family-238 scope page explicitly claims a formalized regular trace estimate and its full-permutation TV consequence covering this source [5]. That is publisher scope metadata, not a Lean execution or proof audit performed here. Its separate conditional-coordinate result distinguishes its feasibility premise from physical-shuffle identification; we do not transfer scope between those statements.

The Araki–Lieb–Thirring inequality is classical. Audenaert's Theorem 1, whose precise statement was inspected, reverses direction at exponent one [4]. For $p/2 \geq 1$, it gives exactly the source's order

$$\operatorname{Tr}|YZ|^p \leq \operatorname{Tr}(Y^*Y)^{p/2}(ZZ^*)^{p/2}. \tag{4}$$

Indeed $|YZ|$ has the relevant nonzero squared spectrum of $(Y^*Y)^{1/2}ZZ^*(Y^*Y)^{1/2}$. We identify no correction to this source step or to its Hölder power argument.

## 2 The source architecture and the genuinely global input

All five cached texts were literally read, including the 633-line scientific proof. The following map separates mechanisms from our independent local results.

1. **Two-orientation smoothing, proof lines 1–29.** For $B = Q^*Q$ or $QQ^*$, an imported routing theorem bounds all $l$-card injection kernels of $B^u$ by $e^{Cl}/(n)_l$. With $l = n$, summing regular diagonals gives $\mathrm{Tr}_{\mathrm{reg}}\,|Q|^{2u} \le e^{Cn}$. A one-card bound alone would not give this full-deck conclusion.

2. **Sparse rows, lines 30–154.** The source polar-deforms blocks of consecutive coordinates. On the imaginary boundary the factors are contractions; on the other boundary both orientations of smoothing enter singular Cauchy–Schwarz. Collision contacts are counted in the actual without-replacement placement law. A connected-component exponential bound and three-lines interpolation give a placement Schatten bound. Branching multiplicities $\binom{q}{k}D_\beta$ then convert it to a small carrier moment. The contacts are not declared independent.

3. **Middle-dimensional overlap, lines 155–540.** For a balanced $a \times b$ grid, row and column carrier rank-one projections act on *all* multiplicity copies. A graded tensor realization records exceptional boxes outside an $r$-hook, $r = \lfloor n^{1/100} \rfloor$. Independent complex Haar planes retain a controlled fraction of the relevant unitary components. Even/odd parities, real carrier vectors, and subgroup factors $|R||F|/(D_R D_F n!)$ are essential. Orthogonal cell tags supply an $\exp(-n)$ compression which cancels the $\exp(n)$ normalization cost. This is not an entrywise small-overlap assertion.

4. **Range coverage and induction, lines 543–609.** Matching invariance kills diagrams of height greater than $n/2$. Large dimensions and sparse rows are covered by fixed powers; the remaining diagrams enter the overlap window. Equation (4) permits positive spectral decompositions on subgroup carriers. The exponent grows by $1 + 2(\log n)^{-1/4}$ at each balanced split. Its product stays bounded along a halving chain of bit lengths. Finite small sizes have a singular gap because the coordinate transpositions generate $S_n$; no explicit size-uniform gap is read off from one finite test.

5. **Physical conversion, lines 611–633.** Hölder converts the trace defect to forward powers and regular row symmetry converts Hilbert–Schmidt norm to TV. The support bound $1 - 2^{tn/2}/n!$ gives the lower order $2d - O(1)$.

The quantitative overlap premise is, with $X = \log D_\lambda$,

$$ne^{-A_0\sqrt{\log n}} \le X \le A_1 n, \qquad D_\lambda \,\mathrm{Tr}(E_R E_F) \le D_R D_F \exp\!\left(\frac{X}{(\log n)^{1/3}}\right).$$

The source's chosen powers of $n$ and logarithms are asymptotic budgets. Our finite refinements below neither establish the imported all-injection routing statement nor make these thresholds effective.

# 3 Regular trace, multiplicity, and heterogeneous sweeps

## 3.1 Exact normalization and the nonnormal distinction

Let $\mu$ be a probability on a finite group $G$, and $K_\mu$ its regular convolution matrix. Every row is a permutation of $(\mu(g))_{g\in G}$, so

$$\|K_\mu - \Pi\|_{\mathrm{HS}}^2 = N \sum_{g\in G} |\mu(g) - N^{-1}|^2, \qquad \mathrm{TV}(\mu, U_G) \le \frac{1}{2} \|K_\mu - \Pi\|_{\mathrm{HS}}. \tag{5}$$

The inequality is Cauchy–Schwarz applied to the $N$ probabilities. No $N^{-1}$ belongs in the regular trace. Relabeling the row proves the same TV estimate from each deterministic initial permutation.

For any integer $v \ge 1$, classical Schatten Hölder gives

$$\|A^v\|_{S_2} \le \|A\|_{S_{2v}}^v .$$

It does not assert $(A^*A)^v = (A^v)^*A^v$. For example, for unit vectors $u, w$ with $c = \langle u, w\rangle \in (0, 1)$, let $P = uu^*$, $R = ww^*$, and $A = PR = cuw^*$. Then

$$\|A^v\|_{\mathrm{HS}}^2 = c^{4v-2}, \qquad \mathrm{Tr}\,|A|^{2v} = c^{2v}.$$

They agree at $v = 1$ and differ at every $v > 1$. This is a diagnostic for implementations, not an error in the source.

**Lemma 3.1** (Multiplicity-sensitive moment conversion)**.** *Suppose $A = \bigoplus_{\lambda\in\Lambda} A_\lambda \otimes I_{D_\lambda}$ on a finite orthogonal direct sum, every $D_\lambda \ge m > 0$, and $\sum_\lambda D_\lambda \mathrm{Tr}\,|A_\lambda|^P \le \varepsilon$, $P > 0$. For every $t \ge P$,*

$$\mathrm{Tr}\,|A|^t \le m^{1-t/P} \varepsilon^{t/P}, \qquad \|A\| \le (\varepsilon/m)^{1/P}. \tag{6}$$

*If the set of blocks is empty, $A = 0$. The estimates are sharp for this abstract block information.*

*Proof.* For each carrier singular value $s_{\lambda j}$, set $x_{\lambda j} = D_\lambda s_{\lambda j}^P$. For $r = t/P \ge 1$,

$$\sum_{\lambda,j} D_\lambda s_{\lambda j}^t = \sum_{\lambda,j} D_\lambda^{1-r} x_{\lambda j}^r \le m^{1-r} \left( \sum_{\lambda,j} x_{\lambda j} \right)^r .$$

The scalar inequality $\sum x_j^r \le (\sum x_j)^r$ follows by dividing by their sum and using $a^r \le a$ for $a \in [0, 1]$. Each $D_\lambda s_{\lambda j}^P \le \varepsilon$ also proves the operator bound. One rank-one carrier block of dimension $m$, with singular value $(\varepsilon/m)^{1/P}$, attains both estimates when $m$ is an available integer dimension. This is algebraic sharpness, not a claim that a sweep realizes that block. □

**Theorem 3.2** (Heterogeneous convolution transfer)**.** *Let $K_i$, $1 \le i \le v$, be independent convolution steps on one finite group, with nontrivial carrier matrices $A_{i,\lambda}$. Let $\Lambda$ be a set of nontrivial carriers containing every nontrivial carrier on which the full product can be nonzero. On its regular sum suppose*

$$\mathrm{Tr}_\Lambda\,|K_i|^{P_i} \le \varepsilon_i, \qquad P_i \ge 2, \qquad S = \sum_{i=1}^v P_i^{-1} \ge \tfrac{1}{2}.$$

*If $\Lambda$ is empty, the product is exactly uniform. Otherwise put $m = \min_{\lambda\in\Lambda} D_\lambda$. Then*

$$\mathrm{TV}(\mu_v * \cdots * \mu_1, U_G) \le \frac{1}{2} m^{1/2-S} \prod_{i=1}^v \varepsilon_i^{1/P_i}. \tag{7}$$

*One may always cap this bound by one. If some $\varepsilon_i = 0$, the product on $\Lambda$ is zero and the conclusion is exact.*

*Proof.* All convolution factors preserve every isotypic summand. We may discard summands on which the product vanishes, including constants after subtracting $\Pi$. Choose $t_i = 2SP_i \geq P_i$; then $\sum_i 1/t_i = 1/2$. Hölder and Lemma 3.1 give

$$\|(K_v - \Pi)\cdots(K_1 - \Pi)\|_{\mathrm{HS}} \leq \prod_i \|K_i|_\Lambda\|_{S_{t_i}} \leq m^{\sum_i 1/t_i - \sum_i 1/P_i} \prod_i \varepsilon_i^{1/P_i}.$$

The product equals $K_v \cdots K_1 - \Pi$. Apply (5). Independence is used to identify the physical product law, not to prove the matrix inequality. □

For equal $P_i = P$ and $\varepsilon_i = \varepsilon$, this yields

$$\mathrm{TV} \leq \tfrac{1}{2} m^{1/2 - v/P} \varepsilon^{v/P} \quad (2v \geq P). \tag{8}$$

Dropping multiplicity gives the familiar $\frac{1}{2}\varepsilon^{v/P}$. At the endpoint $2v = P$ there is no multiplicity gain; it appears only for longer products. The source already retains multiplicities in establishing its moment. Our small increment is their explicit use in this later decay conversion, not the discovery of regular multiplicity.

### 3.2 Which carrier dimensions may be used?

A complete fair coordinate sweep makes one card exactly uniform: successively averaging all coordinates sends a function on $\mathbb{F}_2^d$ to its mean. The natural permutation representation is the direct sum of the trivial and standard representations. Consequently $Q_{(n-1,1)} = 0$. A matching also kills any carrier with no $H_i$-invariant vector. The source, using Pieri, excludes height greater than $n/2$. Retaining this classical exclusion gives the safe finite list

$$\Lambda_n = \{\lambda \vdash n : \ \ell(\lambda) \leq n/2, \ \lambda \neq (n), (n-1,1)\}, \qquad m_n = \min_{\lambda \in \Lambda_n} D_\lambda. \tag{9}$$

The list may still contain further annihilated carriers; that only makes $m_n$ conservative. For $n = 2$ the list is empty and a fair sweep is uniform on $S_2$. No minimum over an empty list is used. For any prescribed coordinate order, the one-card argument and height exclusion are unchanged. Coordinate permutations conjugate the full sweep, preserving its singular moments. Thus (3), if accepted, supplies equal budgets $\varepsilon_i = n^{-10}$ for any independent sequence of such orders.

The hook formula computes $D_\lambda$ exactly from each finite diagram. It gives $m_4 = 2$, $m_8 = 14$, and $m_{16} = 104$. No assertion of an all-$n$ closed form is needed. In particular, blindly inserting $n(n-3)/2$ at $n = 8$ would be wrong: the permitted diagram $(4,4)$ has dimension 14. An illustrative *conditional budget*, not an evaluated sweep, is $P = 4, \varepsilon = 1/64, m = 2, v = 4$; (8) gives $1/(128\sqrt{2})$, compared with $1/128$ after discarding multiplicity. The unspecified source $P$ is not set equal to four by this example.

## 4 Biased matching layers after a smoothed prefix

This section perturbs the actual pair-switching law, not an arbitrary Gaussian surrogate. For a fixed coordinate matching with edges $e$, let $U_e$ be its commuting self-adjoint involutions. Independent switch probabilities $p_e \in [0,1]$ give

$$B = \prod_e [(1-p_e)I + p_e U_e], \qquad P = \prod_e \tfrac{1}{2}(I + U_e), \qquad \rho = \max_e |1 - 2p_e|. \tag{10}$$

Both $B$ and $P$ are regular convolution Markov contractions.

**Proposition 4.1** (A layer budget independent of pair count)**.** *In every unitary representation, $\|B - P\| \le \rho$. For a full sweep with layer budgets $\rho_1, \dots, \rho_d$,*

$$\left\|\widetilde{Q} - Q\right\| \le \eta := \sum_{i=1}^{d} \rho_i. \tag{11}$$

*The matching estimate is sharp in the regular representation whenever there is an edge attaining $\rho$.*

*Proof.* Simultaneous diagonalization of the commuting involutions labels a joint eigenspace by a subset $F$ of edges with eigenvalue $-1$. On $F = \varnothing$, $B = P = I$; otherwise $P = 0$ and $B$ has eigenvalue $\prod_{e\in F}(1 - 2p_e)$, whose absolute value is at most $\rho$. The regular representation restricted to the matching subgroup contains every character: functions on each subgroup coset give its regular module. The singleton character for a maximizing edge therefore attains the bound. Finally expand the difference of the two ordered sweep products by telescoping; every surrounding layer has norm at most one. □

This is substantially smaller than adding the absolute biases of all $nd/2$ coins when the error criterion is operator norm. It is *not* a dimension-free trace bound for a wholly biased shuffle. A small operator error may occupy many regular directions. The following theorem states exactly where smoothing makes the operator error sufficient.

**Theorem 4.2** (Post-smoothing perturbation ledger)**.** *Let $H$ be a convolution Markov operator with $\|H - \Pi\|_{\mathrm{HS}} \le h$. Append $L$ ideal convolution Markov steps $K_1, \dots, K_L$, or approximate steps $\widetilde{K}_1, \dots, \widetilde{K}_L$, in that order. On constants' orthogonal complement assume*

$$\|K_j\| \le q_j, \quad \left\|\widetilde{K}_j\right\| \le \widetilde{q}_j, \quad \left\|\widetilde{K}_j - K_j\right\| \le \eta_j, \qquad 0 \le q_j, \widetilde{q}_j \le 1.$$

*Then the approximate law satisfies both bounds*

$$2\,\mathrm{TV}(\widetilde{K}_L \cdots \widetilde{K}_1 H, U_G) \le h \prod_{j=1}^{L} \widetilde{q}_j, \tag{12}$$

$$2\,\mathrm{TV}(\widetilde{K}_L \cdots \widetilde{K}_1 H, U_G) \le h \left[\prod_{j=1}^{L} q_j + \sum_{j=1}^{L} \eta_j \prod_{i=j+1}^{L} \widetilde{q}_i \prod_{i=1}^{j-1} q_i\right]. \tag{13}$$

*In particular, the TV distance between approximate and ideal laws is bounded by half the displayed sum times $h$.*

*Proof.* Restrict to constants' orthogonal complement and put $H_0 = H - \Pi$. The ideal and approximate errors are the respective products applied to $H_0$. The ideal-property inequality $\|XYZ\|_{\mathrm{HS}} \le \|X\|\,\|Y\|_{\mathrm{HS}}\,\|Z\|$ proves (12). The exact ordered identity is

$$\widetilde{K}_L \cdots \widetilde{K}_1 - K_L \cdots K_1 = \sum_{j=1}^{L} \widetilde{K}_L \cdots \widetilde{K}_{j+1} (\widetilde{K}_j - K_j) K_{j-1} \cdots K_1.$$

Multiply on the right by $H_0$, take Hilbert–Schmidt norms, and use (5), also for the difference of two convolution laws. Adding the ideal bound gives (13). □

For equal budgets $q_j = q$, $\eta_j = \eta$, one can use $\widetilde{q} = \min(1, q + \eta)$. When $q + \eta \le 1$, the bracket in (13) is exactly $(q + \eta)^L$; this includes $q = 0$ and $\eta = 0$ by the polynomial binomial identity. Alternatively keep a measured smaller $\widetilde{q}$ in the finite sum. If all $q_j = 1$, the coarse bound is $h(1 + \sum_j \eta_j)$, while the direct contraction estimate remains $h$; the sum is then useful for comparing the two laws, not for improving distance to uniform.

### 4.1 What a source-linked certificate would contain

Here assume $n \geq 4$; for $n = 2$, one fair sweep is already uniform and subsequent convolution steps preserve that law. Take an exact prefix of $k$ fair sweeps, $2k \geq P$. Theorem 3.2 gives

$$h \leq m_n^{1/2-k/P} n^{-10k/P}.$$

For an appended ideal fair sweep, $q \leq \min(1, (n^{-10}/m_n)^{1/P})$. For its independent biased layers, Proposition 4.1 gives $\eta = \sum_i \max_e |1 - 2p_{i,e}|$. All quantities except the imported source power $P$ are explicit once $n$ and the biases are specified.

As a budget illustration, let $h = 1/100$, $q = 1/4$, two layer biases each at most $1/32$ in absolute probability, and $L = 3$. Each $\rho_i \leq 1/16$, hence $\eta \leq 1/8$ and $\widetilde{q} \leq 3/8$. The output TV is at most

$$\frac{1}{200}\left(\frac{3}{8}\right)^3 = \frac{27}{102400}.$$

This is a rational consequence of supplied budgets, not evidence that those budgets hold for a particular unspecified large deck. Errors *before* the smoothed prefix, biased replacements of every layer, dependence between coins, or a state-adaptive choice of sweep order require new hypotheses. The theorem does not cover them by renaming the prefix.

## 5 A finite refinement of the cell-tag compression

We next retain a piece of exact orthogonality which the source discards in a convenient coarse bound. The exponent order of its global argument is unchanged.

Let $D$ be a finite-dimensional $\mathbb{Z}/2\mathbb{Z}$-graded Hilbert space, with mutually orthogonal parity-stable tag subspaces $D_1, \ldots, D_n$. Let $e$ be any unit vector in $D_1 \otimes \cdots \otimes D_n$, possibly entangled. Let $W \subset D$ be an even subspace of dimension $w \geq 1$, and let $P_*$ be the orthogonal projection onto graded symmetric tensors. For $S \subset \{1, \ldots, n\}$, let $\Delta_S$ test that exactly the slots in $S$ lie in $W^\perp$. Put $\Delta_t = \sum_{|S|=t} \Delta_S$, and $\Delta_{\leq T} = \sum_{t=0}^{T} \Delta_t$, where $0 \leq T < n$. For $m = n - t > 0$, define

$$c_t(w) = \frac{\binom{m+w-1}{w-1}}{t!\, m^m}, \qquad C_{\text{count}} = n! \sum_{t=0}^{T} \binom{n}{t}^2 c_t(w). \tag{14}$$

**Proposition 5.1** (Orthogonal exception-count budget)**.** *In the preceding setting,*

$$n!\, \|P_* \Delta_{\leq T} e\|^2 \leq C_{\text{count}} \leq n! \left(\sum_{t=0}^{T} \binom{n}{t}\right)^2 \max_{0 \leq t \leq T} c_t(w). \tag{15}$$

*The last quantity is the form of the source's cell-compression bound. One can also cap the first bound by $n!$.*

*Proof.* We first reproduce the fixed-assignment estimate to identify its hypotheses. Group the $m$ slots outside $S$ and the $t$ slots in $S$, retaining graded reordering signs. Full invariant projection is dominated by within-group invariant projection. The latter commutes with $\Delta_S$. After relaxing the outside test, the relevant operator is dominated by

$$P_{\mathrm{Sym}^m W} \otimes P_{\text{graded symmetric},t}.$$

Compress it to the product tag subspace. On the last $t$ slots, any nonidentity permutation changes at least one orthogonal tag; the compressed second factor is exactly $I/t!$. For the first factor, unitary invariance on $W$ gives the classical coherent-orbit identity

$$P_{\mathrm{Sym}^m W} = \binom{m+w-1}{w-1} \mathbb{E}_z \, |z^{\otimes m}\rangle\langle z^{\otimes m}|,$$

where $z$ is uniform on the complex unit sphere in $W$. The integral commutes with $U(W)$, is scalar on the irreducible symmetric power, and has trace one before multiplication by its dimension; these facts prove the identity. If $C_i$ is the tag projection in a retained slot, the compressed rank-one norm is $\prod_i \|C_i z\|^2 \le m^{-m}$. Indeed the $m$ nonnegative factors have sum at most one, by tag orthogonality, and arithmetic–geometric mean applies. Thus $\|P_* \Delta_S e\|^2 \le c_t(w)$, even for entangled $e$.

For fixed $t$, triangle inequality and Cauchy–Schwarz give

$$\|P_* \Delta_t e\|^2 \le \binom{n}{t} \sum_{|S|=t} \|P_* \Delta_S e\|^2 \le \binom{n}{t}^2 c_t(w).$$

The crucial retained fact is that each $\Delta_t$ commutes with every graded slot permutation: it counts exceptions without selecting their positions. Therefore $P_* \Delta_t e$ and $P_* \Delta_u e$ are orthogonal for $t \neq u$. Summing their squared norms proves the first inequality. The second follows from $\sum_t \binom{n}{t}^2 \le (\sum_t \binom{n}{t})^2$. Finally $P_* \Delta_{\le T}$ is a projection, giving the cap. □

This proof does not claim that individual $P_* \Delta_S e$ are orthogonal. Slot symmetrization can make them parallel. Only different *counts* are separated. For $n = 4, T = 1, w = 1$, the two uncapped certificates are

$$C_{\mathrm{count}} = 24 \left( \frac{1}{256} + \frac{16}{27} \right) = \frac{4123}{288}, \qquad C_{\mathrm{source}} = \frac{200}{9}.$$

Both remain below the trivial cap 24, so the strict improvement survives capping. These are finite bounds, not values of the tested tensor norm.

For the source, $w \le 4r^2$ and $T = L_A + L_B + 4r^2 \le CX/\log n$. The replacement (14) has only $T + 1$ scalar summands. It preserves the same subgroup normalization and Haar retention factors. It does not remove the two factors of $\binom{n}{t}$ inside a count, and it still has the source's $-n + O(T \log(en/T) + r^2 \log n)$ logarithmic bound. Hence there is no claimed new overlap exponent, induction power, or global mixing constant. Its value is an exact smaller finite upper bound when those source parameters are already certified.

## 6 Physical metrics and representation tails

The original private physical-metric proofs [6, 8] explain why nonorthogonal coefficients must carry their Gram matrix. If $J : \mathbb{C}^D \to V$ is an isomorphism onto an entire carrier, $G = J^* J > 0$, and $QJ = JB$, then the correct orthonormal matrix is

$$B_{\mathrm{phys}} = G^{1/2} B G^{-1/2}, \qquad B^{\dagger_G} = G^{-1} B^* G. \tag{16}$$

In particular, coordinate conditional projections obey $B_i^2 = B_i$ and $B_i^{\dagger_G} = B_i$, not necessarily $B_i^* = B_i$ in the raw Euclidean coordinates. Trace is similarity-invariant; singular-value trace is not.

For example, let

$$A = \begin{pmatrix} 0 & 1/2 \\ 0 & 0 \end{pmatrix}, \quad J = \mathrm{diag}(1, t), \quad G = \mathrm{diag}(1, t^2), \quad B = J^{-1} A J = \begin{pmatrix} 0 & t/2 \\ 0 & 0 \end{pmatrix}.$$

The raw nonzero singular value is $t/2$, but the physical one is $1/2$, for every $t > 0$. This counterexample also has identically zero eigenvalues, so eigenvalue radii provide no replacement for physical singular values. The later comparison report [9] gives a related Jordan resolvent diagnostic; it is credited as a comparison, not as the original physical proof.

**Proposition 6.1** (A full-carrier moment certificate)**.** *Fix $P \geq 2$. For each retained nontrivial carrier $\lambda$, suppose its* full *coordinate matrix is $B_\lambda$, with exact Gram $G_\lambda > 0$. Suppose*

$$a_\lambda G_\lambda \preceq \widehat{G}_\lambda \preceq b_\lambda G_\lambda, \qquad \left\|B_\lambda - \widehat{B}_\lambda\right\|_{\widehat{G}_\lambda} \leq e_\lambda, \qquad 0 < a_\lambda \leq b_\lambda,$$

*and $\left\|\widehat{B}_\lambda\right\|_{S_P,\widehat{G}_\lambda} \leq z_\lambda$. If the omitted regular moment is at most $E_{\mathrm{tail}}$, then the total nonconstant regular moment is at most*

$$E_{\mathrm{tail}} + \sum_{\lambda \text{ retained}} D_\lambda \left[\sqrt{\frac{b_\lambda}{a_\lambda}}\,(z_\lambda + D_\lambda^{1/P} e_\lambda)\right]^P . \tag{17}$$

*Proof.* The two orthonormal coordinate realizations are conjugate by $L = G_\lambda^{1/2}\widehat{G}_\lambda^{-1/2}$. Its condition number is at most $\sqrt{b_\lambda/a_\lambda}$ by the assumed quadratic-form inequalities. The ideal-property bound for Schatten norms gives $\|B\|_{S_P,G} \leq \sqrt{b/a}\,\|B\|_{S_P,\widehat{G}}$. A rank at most $D_\lambda$ operator of norm $e_\lambda$ has $S_P$-norm at most $D_\lambda^{1/P} e_\lambda$. Use the Schatten triangle inequality, raise to $P$, multiply by regular multiplicity, and add the declared tail. □

Positive rational $LDL^*$ factorizations and interval residuals can certify the finite Gram inequalities, as in the fresh original TR248 passage. They do not certify $E_{\mathrm{tail}}$. If the matrices describe only a compressed subspace rather than the full carrier, $QJ = JB$ must be replaced by an equation with an omitted component. The original physical residual identity is

$$\|(Q - \lambda)Jc\|^2 = \|(B - \lambda)c\|_G^2 + \|(I - P_J)QJc\|^2 .$$

For the present task the needed tail is a regular Schatten moment, not merely this single-vector leakage. The private semigroup proof [7] independently illustrates the distinction: operator-norm convergence can hold while traces diverge or have a nonvanishing normalized bias. We use that lesson, not its self-adjoint spectral theorem, for a nonnormal sweep.

# 7 Worked checks, advantages, and limitations

The owned script `_audit/M473-checks.py` uses exact rational arithmetic and finite permutations. It constructs the genuine $S_4$ coordinate-matching laws, with independent edge coins, and compares their convolution products. Regular Hilbert–Schmidt norms are evaluated as 24 times squared probability-density error, not by normalizing the trace by 24. Even Schatten moments are computed from convolution with the reversed law. It also enumerates finite partitions for the three hook minima, checks scalar multiplicity inequalities, matching-character budgets, ordered perturbation identities, nonnormal rank-one examples, and the exception-count formulas. For the actual two $S_4$ coordinate orders, the computed nonconstant moments at $P = 2, 4, 6, 8$ are respectively $1/2, 1/8, 1/32, 1/128$. This is a finite exact calculation, not the much smaller $n^{-10}$ budget at the source's unspecified larger power. The script also directly projects the even tag tensors for $n = 2, 3, 4$ and checks orthogonality of distinct exception-count sectors.

Every test group records parameter cases separately from assertions. These finite fixtures test our implementation and the printed local algebra. They do not test every source Haar-plane argument, the companion routing theorem, all deck sizes, or a Lean development. The final run passed 1,612 cases and 3,782 assertions, with counts and case definitions stored in the test receipt and readiness record. No parent's tests or foreign checker replay enters them.

**Advantages.** Multiplicity can improve a longer-product estimate without modifying the physical shuffle. Different prescribed coordinate orders need not commute. The matching-bias budget counts the worst edge bias per layer, not the number of coin flips, in its correct operator norm. A smoothed prefix makes that budget usable without a new factor of $\sqrt{n!}$. The exception-count refinement is a finite sum and retains all parity and subgroup conventions.

**Costs and limitations.** The exact source power $P$, asymptotic thresholds, and companion constants are not supplied by these refinements. Hook enumeration and full-carrier matrix construction can be expensive; no measured speedup or polynomial-time implementation is claimed. The post-smoothing theorem requires the prefix to be certified under its actual law. Biased layers need independent pair coins, and the full scalar maximum may overestimate the active spectrum inside an individual carrier. Entrywise matrix tolerances alone do not provide the metric or tail certificate in (17).

**Extensions requiring additional work.** A useful next step is to bound omitted carrier moments using the source's dimension ranges while evaluating only borderline blocks. This needs effective range constants. A second is to optimize heterogeneous exponents and retain carrier-specific rather than minimum multiplicities; the resulting bound must still sum every regular copy. A third is to exploit geometry of particular retained Haar planes in the cell-tag integral, rather than the universal arithmetic–geometric-mean estimate. Finally, a wholly biased shuffle needs a trace argument from time zero; an operator-small perturbation of fair sweeps alone does not prove it.

## 8 Attribution and evidence boundary

Applicability remains L3. The contribution is assessed C2, because the increments are finite refinements assembled from classical Schatten Hölder, scalar power inequalities, commuting involutions, orthogonal projections, and physical Gram transport. They are useful for a source-specific certificate but do not constitute a new proof of optimal-order Thorp mixing or a substantial new representation theorem.

The public read comprises all 875 lines in the five cached text files. Fresh private reading comprises the full 210-line physical metric chapter, 242-line spectral-limit chapter, 261-line semigroup module, and original TR248 lines 408–478 and 1139–1268 (201 lines); the later comparison was read at lines 190–281 (92 lines), separately. Thus 914 original private lines and 92 comparison lines were read, not the whole private corpus. Included files and all external cited proofs not individually read remain outside this scope. Pre-read and post-read hashes and the unread complements are retained in the owned ledger.

No scientific error was identified in the source's Hölder, ALT ordering, regular multiplicity, or two-orientation normalization. Our exception-count inequality strengthens a deliberately coarse source bound; it is not a repair of a false theorem. Captured source-tree byte identity is provenance, not a claim about a live branch or mathematical validity. The public primary refresh is dated October 9, 2026; no confidential text or title was uploaded.

## References


[1] OpenAI, *Random-subspace tests and trace smoothing for coordinate sweeps*, September 26, 2026, cached assigned manuscript. Public source directory. Fresh full scientific text read; companion inputs remain imported.

[2] OpenAI, *Routing densities and representation contraction for Thorp sweeps*, September 26, 2026. The assigned source imports Theorem 6.1 and Corollary 6.2; their full proofs were not read in this assignment.

[3] B. Morris, *Improved mixing time bounds for the Thorp shuffle*, Combinatorics, Probability and Computing **22** (2013), 118–132. Publisher DOI; arXiv:0912.2759v1. Primary abstract and metadata refreshed, not a full proof audit.

[4] K. M. R. Audenaert, *On the Araki–Lieb–Thirring inequality*, International Journal of Information and Systems Sciences **4** (2008), 78–83. arXiv:math/0701129v2, Theorem 1 and the displayed Hölder statement in Section 2.

[5] OpenAI, *Optimal logarithmic mixing of the Thorp shuffle*, family-238 formalization scope page, accessed October 9, 2026. Publisher metadata only; no Lean execution.

[6] M. Subbarao, original private RamCalBooks, volume 21, `chapters/ch01d-physical-metrics.tex`, lines 1–210; `chapters/ch01f-spectral-limits.tex`, lines 1–242. Physical Gram, omitted residual, metric approximation, and finite-to-global limitations. Confidential local originals.

[7] M. Subbarao, original private RamCalBooks, volume 21, `chapters/ch03/semigroup.tex`, lines 1–261. Resolvent time stepping, trace costs, and full error budgets. Confidential local original; self-adjoint conclusions not transplanted to the nonnormal shuffle.

[8] M. Subbarao, original private `RamFoundationTRs/TR-rev-1-old-tr248/TR-rev-1-old-tr248.tex`, lines 408–478 and 1139–1268. Physical coordinates, moment Gram assembly, and rigorous margins.

[9] M. Subbarao, private later comparison `res-math/TR-math19-rev2.tex`, lines 190–281. Nonnormal resolvent and Jordan diagnostics. Used as a comparison, not as a substitute for original Ram proofs.